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MATHEMATICS TEST - CHAPTER 10: PRACTICAL GEOMETRY

Name:		Roll No.		Section:

Total Time: 2:30 Hour **Total Marks: 150**
Time Allowed: 150 Minutes

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. Which instrument is used to measure an angle?	Compass	Ruler	Divider	Protractor	○○○●
2. Which instrument is used to draw an arc of a given radius?	Protractor	Ruler	Compass	Set square	○○●○
3. Which instrument is needed to draw AB = 6 cm?	Compass alone	Protractor alone	Ruler	Pencil alone	○○●○
4. To mark ∠XYA = 50° at Y, which instrument is most direct?	Protractor	Compass alone	Ruler	Divider	●○○○
5. The sharp point of a compass is placed at the	Centre of the arc	End of the arc	Edge of the ruler	Pencil mark	●○○○
6. A semicircular protractor can measure angles up to	90°	180°	270°	360°	○●○○
7. To construct 60° with compass and ruler, arcs are drawn with	Radii in ratio 2:1	Radii 1 cm and 2 cm	Unequal radii	Equal radii	○○○●
8. Angles on a protractor are measured in	Centimetres	Square units	Grams	Degrees	○○○●
9. An unmarked edge used only to draw straight lines is a	Straightedge	Protractor	Compass	Divider	●○○○
10. A compass keeps its radius fixed, so every point of the arc is	Collinear	Equidistant from the centre	On the ruler edge	At 90° to the centre	○●○○
11. Triangle from two sides and the included angle: which case?	SSS	SAS	ASA	RHS	○●○○
12. In the SAS case, the given angle lies	Opposite the longer side	Between the two given sides	Opposite the shorter side	Outside the triangle	○●○○
13. SAS: AB = 5 cm, BC = 4 cm, ∠B = 60°. Step after drawing AB?	Make ∠ABX = 60° at B	Join AC	Cut an arc of 4 cm at A	Bisect AB	●○○○
14. SAS: after making ∠ABX = 60° at B, the next step is to	Cut BC = 4 cm on BX	Draw an altitude	Bisect ∠ABX	Draw a median	●○○○

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Questions	A	B	C	D	A B C D
15. The last step in the SAS construction of △ABC is to	Draw ∠C	Bisect AC	Extend BA	Join A and C	○○○●
16. SAS: AB = 5, BC = 4, ∠B = 60°. Vertex C lies on the ray from B making 60° with	AC	Median of AB	Bisector of AB	BA	○○○●
17. Which data fits the SAS case for △PQR?	PQ = 4, QR = 6, ∠P = 70°	PQ = 4, PR = 6, ∠Q = 70°	∠P = 40°, ∠Q = 70°, ∠R = 70°	PQ = 4, QR = 6, ∠Q = 70°	○○○●
18. AB = BC = 6 cm and ∠B = 60° are constructed. △ABC is	Right-angled	Scalene	Obtuse	Equilateral	○○○●
19. AB = BC = 5 cm, ∠B = 90° are constructed. Then AC =	10 cm	5√2 cm	5 cm	25 cm	○●○○
20. SAS: AB = 3 cm, BC = 4 cm, ∠B = 90°. Length of AC is	5 cm	7 cm	6 cm	25 cm	●○○○
21. AB = AC = 6 cm and ∠A = 90° give a triangle that is	Equilateral	Scalene acute	Obtuse isosceles	Right isosceles	○○○●
22. SAS: AB = AC = 4 cm, ∠A = 100°. Each base angle is	80°	50°	40°	100°	○○●○
23. If SAS data has the included angle ∠B = 120°, △ABC is	Obtuse-angled	Acute-angled	Right-angled	Equilateral	●○○○
24. How many triangles (up to congruence) exist for given SAS data?	Two	None	Infinitely many	One	○○○●
25. SAS: AB = 8 cm, BC = 6 cm, ∠B = 90°. Perimeter of △ABC is	14 cm	28 cm	48 cm	24 cm	○○○●
26. Constructing a triangle from one side and two angles is the	SAS case	ASA case	SSS case	RHS case	○●○○
27. XY = 6 cm, ∠X = 45°, ∠Y = 60°. The angle ∠Z is	105°	75°	65°	85°	○●○○
28. In Example 7, ∠XYB = 60° is constructed at the point	Y	X	Z	Midpoint of XY	●○○○
29. In Example 7, vertex Z is the point where	Rays XA and YB meet	Arcs of XY meet	XY meets XA	Altitudes meet	●○○○
30. In ASA, the given side lies	Opposite the larger angle	Opposite the smaller angle	Between the two given angles	Outside the triangle	○○●○
31. ASA data ∠A = 50° and ∠B = 70° give a triangle with ∠C =	50°	70°	60°	80°	○○●○

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Questions	A	B	C	D	A B C D
32. Why can angles of 100° and 90° not be in one triangle?	Their sum exceeds 180°	Their sum is below 180°	Both are obtuse	Both are right angles	●○○○
33. Base BC = 5 cm with ∠B = ∠C = 60° gives a triangle that is	Right-angled	Obtuse isosceles	Equilateral	Scalene	○○●○
34. ASA: BC = 7 cm, ∠B = ∠C = 45°. Then ∠A is	45°	90°	60°	100°	○●○○
35. ASA: ∠A = 30° and ∠B = 60°. The triangle is	Equilateral	Obtuse	Right-angled	Isosceles obtuse	○○●○
36. Which pair cannot be two angles of a triangle?	80° and 90°	110° and 70°	60° and 60°	30° and 120°	○●○○
37. To construct BC = 5, ∠B = 40°, ∠C = 70°, the first step is to	Draw BC = 5 cm	Draw ∠A	Bisect BC	Draw a circle	●○○○
38. In that ASA construction, vertex A is found where	Rays from B and C meet	Arcs of radius 5 meet	BC meets its bisector	Altitude meets BC	●○○○
39. Two angles of a triangle are 35° and 65°. The third is	90°	100°	80°	70°	○○●○
40. ASA: ∠B = ∠C = 50°, BC = 6 cm. Which is true?	AB = BC	AB = AC	AC = BC	AB > BC	○●○○
41. Two sides and an angle opposite to one of them: which case?	SSA	SAS	ASA	SSS	●○○○
42. Right-angled triangle from hypotenuse and one side is the	SAS case	ASA case	AAA case	RHS case	○○○●
43. In SSA, the given angle is	Between the two sides	Outside the triangle	Always 90°	Opposite one of the given sides	○○○●
44. SSA: ∠A = 40°, AB = 6, BC = 4. The arc of radius 4 is drawn from	A	Midpoint of AB	C	B	○○○●
45. In that construction, the arc of radius 4 cuts the ray from A at	A	C	B	Midpoint of AB	○●○○
46. Right triangle, hypotenuse 5 cm, one side 3 cm. Other side is	2 cm	4 cm	8 cm	34 cm	○●○○
47. Right triangle, hypotenuse 13 cm, one side 5 cm. Other side is	8 cm	18 cm	12 cm	144 cm	○○●○
48. In the RHS case, the right angle is opposite to the	Shortest side	Hypotenuse	Base	Median	○●○○

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Questions	A	B	C	D	A B C D
49. In SSA, if the arc does not meet the ray, then	A unique triangle forms	No triangle is formed	Two triangles form	An equilateral forms	○●○○
50. In SSA, if the arc cuts the ray at two points, there are	One triangle	No triangle	Two triangles	Three triangles	○○●○
51. In SSA, if the arc just touches the ray, the triangle is	Right-angled	Equilateral	Obtuse isosceles	Not possible	●○○○
52. Right triangle with hypotenuse 10 cm and one side 6 cm: third side	4 cm	16 cm	8 cm	12 cm	○○●○
53. In a right triangle, the hypotenuse is	The longest side	The shortest side	A median	An altitude	●○○○
54. Right angle at B, BC = 3, AC = 5. The arc of radius 5 is drawn from	B	A	Midpoint of BC	C	○○○●
55. Isosceles right triangle with hypotenuse 10 cm has each leg	5 cm	10 cm	$10\sqrt{2}$ cm	$5\sqrt{2}$ cm	○○○●
56. Sides 3 cm, 4 cm, 8 cm: can a triangle be constructed?	Yes, scalene	No, 3 + 4 < 8	Yes, right-angled	Yes, obtuse	○●○○
57. Which set of lengths can be sides of a triangle?	2, 3, 6	4, 4, 9	1, 2, 3	5, 7, 10	○○○●
58. The sum of any two sides of a triangle is always … the third	Less than	Equal to	Greater than	Double	○○●○
59. Two sides are 5 cm and 8 cm. The third side x satisfies	$5 < x < 8$	$3 < x < 13$	$x > 13$	$x < 3$	○●○○
60. Two sides are 7 cm each. Greatest whole-number third side is	13 cm	14 cm	12 cm	7 cm	●○○○
61. A triangle with sides 6, 8, 10 is	Acute-angled	Obtuse-angled	Right-angled	Equilateral	○○●○
62. A triangle with sides 5, 5, 8 is	Isosceles	Equilateral	Scalene	Right-angled	●○○○
63. Sides 2 cm, 3 cm, 5 cm give	Isosceles triangle	No triangle (collinear)	Scalene triangle	Right triangle	○●○○
64. Perimeter of a triangle with sides 7, 8, 9 cm is	16 cm	25 cm	24 cm	22 cm	○○●○
65. Smallest whole number x so that 3, 5, x form a triangle	2	3	8	1	○●○○

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Questions	A	B	C	D	A B C D
66. Largest whole number x so that 3, 5, x form a triangle	8	15	7	2	○○●○
67. Can 50°, 60° and 80° be the angles of a triangle?	No, sum is 190°	Yes, acute	Yes, scalene	No, sum is 170°	●○○○
68. Angles 45°, 45°, 90° form a triangle that is	Equilateral	Obtuse	Right isosceles	Scalene acute	○○●○
69. The sum of the interior angles of a triangle is	180°	360°	90°	270°	●○○○
70. A triangle can have at most how many obtuse angles?	One	Two	Three	None	●○○○
71. The longest side of a triangle is opposite the	Smallest angle	Right angle only	Largest angle	Middle angle	○○●○
72. In △ABC, ∠A = 70°, ∠B = 60°. Which side is longest?	AB	BC	AC	All equal	○●○○
73. In △ABC, ∠A = 50° and ∠B = 60°. Exterior angle at C is	70°	110°	130°	120°	○●○○
74. Perimeter of an equilateral triangle of side 6 cm is	12 cm	18 cm	36 cm	6√3 cm	○●○○
75. Area of a triangle with sides 3, 4, 5 cm is	12 cm²	7.5 cm²	10 cm²	6 cm²	○○○●
76. A perpendicular from a vertex to the opposite side is called	Altitude	Median	Angle bisector	Perpendicular bisector	●○○○
77. The altitudes of a triangle are	Parallel	Equal in length	Concurrent	Collinear	○○●○
78. The point where the three altitudes meet is the	Orthocentre	Centroid	Incentre	Circumcentre	●○○○
79. The altitudes of an acute-angled triangle meet	Outside the triangle	At a vertex	Inside the triangle	At a side midpoint	○○●○
80. The altitudes of a right-angled triangle meet at	Midpoint of hypotenuse	A point outside	The right-angle vertex	The centroid	○○●○
81. The altitudes of an obtuse-angled triangle meet	Inside the triangle	At a vertex	On the longest side	Outside the triangle	○○○●
82. The altitude from C makes with AB an angle of	60°	45°	90°	180°	○○●○

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Questions	A	B	C	D	A B C D
83. How many altitudes does a triangle have?	3	2	1	6	●○○○
84. In Example 7, altitude XD is drawn so that XD ⊥	XY	XZ	YZ	ZC	○○●○
85. In an equilateral triangle, an altitude is also a	Median only	Side	Hypotenuse	Median and angle bisector	○○○●
86. Legs 3 cm and 4 cm: altitude to the hypotenuse is	2.4 cm	2.5 cm	3.5 cm	1.2 cm	●○○○
87. Altitude of an equilateral triangle of side 6 cm is	$3\sqrt{3}$ cm	$6\sqrt{3}$ cm	3 cm	9 cm	●○○○
88. A triangle has base 10 cm and altitude 4 cm. Its area is	40 cm^2	20 cm^2	14 cm^2	10 cm^2	○●○○
89. A line joining a vertex to the midpoint of the opposite side is a	Altitude	Angle bisector	Perpendicular bisector	Median	○○○●
90. The three medians of a triangle meet at the	Orthocentre	Incentre	Circumcentre	Centroid	○○○●
91. The centroid of a triangle lies	Outside an obtuse one	On the hypotenuse of a right one	At a vertex	Always inside it	○○○●
92. The centroid divides each median in the ratio (from vertex)	1 : 2	1 : 1	3 : 1	2 : 1	○○○●
93. A median is 12 cm long. Centroid is how far from the vertex?	4 cm	8 cm	6 cm	9 cm	○●○○
94. A median is 9 cm long. Centroid is how far from the midpoint?	6 cm	3 cm	4.5 cm	9 cm	○●○○
95. To draw a median, we first locate the	Foot of a perpendicular	Midpoint of the opposite side	Bisector of an angle	Circumcentre	○●○○
96. The midpoint of a side is found by drawing its	Angle bisector	Perpendicular bisector	Altitude	Parallel line	○●○○
97. A median of a triangle divides its area into	Parts in ratio 2:1	Parts in ratio 1:3	Two equal parts	Two similar triangles	○○●○
98. D is the midpoint of BC and BC = 10 cm. Then BD =	10 cm	5 cm	20 cm	2.5 cm	○●○○
99. How many medians does a triangle have?	2	1	6	3	○○○●

Questions	A	B	C	D	A B C D
100. In isosceles △ABC (AB = AC), the median from A is also an	Side	Hypotenuse	Parallel to BC	Altitude	○○○●
101. In a right triangle, the median to the hypotenuse equals	The hypotenuse	A third of it	The shorter leg	Half the hypotenuse	○○○●
102. Hypotenuse of a right triangle is 10 cm. Median to it is	10 cm	5 cm	2.5 cm	20 cm	○●○○
103. A ray that divides an angle into two equal parts is its	Median	Altitude	Perpendicular	Bisector	○○○●
104. The angle bisectors of a triangle meet at the	Incentre	Centroid	Orthocentre	Circumcentre	●○○○
105. The incentre is equidistant from the	Three vertices	Three midpoints	Two vertices only	Three sides	○○○●
106. The circle touching all three sides of a triangle is the	Circumcircle	Excircle	Incircle	Unit circle	○○●○
107. The angle bisectors of a triangle meet	Always inside it	Outside an obtuse one	At the right vertex	On a side	●○○○
108. To bisect ∠ABC, the first arc is drawn with centre	A	B	C	Midpoint of AC	○●○○
109. In bisecting an angle, equal arcs are then drawn from	The two points cut on the arms	The vertex again	Midpoints of the arms	Any point inside	●○○○
110. The bisector of an 80° angle makes angles of	80° each	40° each	20° each	160° each	○●○○
111. BD bisects ∠ABC = 70°. Then ∠ABD =	70°	140°	35°	30°	○○●○
112. Bisector of ∠C = 40° of a triangle makes angles of	40° each	20° each	30° each	50° each	○●○○
113. The radius of the incircle is the distance from incentre to	A side	A vertex	The centroid	The orthocentre	●○○○
114. Minimum number of angle bisectors needed to locate the incentre	One	Three	Two	Four	○○●○
115. Bisecting a 90° angle twice gives an angle of	45°	30°	22.5°	15°	○○●○
116. A line perpendicular to a segment through its midpoint is its	Median	Altitude	Angle bisector	Perpendicular bisector	○○○●

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Questions	A	B	C	D	A B C D
117. The perpendicular bisectors of the sides meet at the	Circumcentre	Incentre	Centroid	Orthocentre	●○○○
118. The circumcentre is equidistant from the	Three sides	Three vertices	Three medians	Two sides only	○●○○
119. To bisect AB perpendicularly, arcs from A and B use a radius	Less than half of AB	Exactly AB/3	More than half of AB	Zero	○○●○
120. Equal arcs from A and B cut each other at	One point on AB	Two points, which are joined	The midpoint directly	Point C	○●○○
121. Any point on the perpendicular bisector of AB is	Nearer to A	Nearer to B	Equidistant from A and B	On AB only	○○●○
122. The circumcentre of an acute-angled triangle lies	Inside the triangle	Outside the triangle	On a side	At a vertex	●○○○
123. The circumcentre of a right-angled triangle lies at the	Right-angle vertex	Midpoint of the hypotenuse	Centroid	Point outside	○●○○
124. The circumcentre of an obtuse-angled triangle lies	Outside the triangle	Inside the triangle	On a side	At a vertex	●○○○
125. The circle passing through all three vertices is the	Incircle	Excircle	Circumcircle	Semicircle	○○●○
126. A right triangle has legs 6 cm and 8 cm. Circumradius is	10 cm	4 cm	5 cm	7 cm	○○●○
127. A segment of 8 cm is bisected perpendicularly. Each part is	4 cm	8 cm	2 cm	16 cm	●○○○
128. A perpendicular bisector meets the segment at an angle of	45°	60°	180°	90°	○○○●
129. Lines passing through the same point are called	Parallel	Collinear	Concurrent	Congruent	○○●○
130. Which pair of points always lies inside every triangle?	Incentre and orthocentre	Centroid and circumcentre	Incentre and centroid	Orthocentre and circumcentre	○○●○
131. In an equilateral triangle, the four centres	Coincide	Lie on the vertices	Lie outside	Are all different	●○○○
132. Which construction uses a perpendicular from a vertex?	Median	Angle bisector	Altitude	Side bisector	○○●○
133. Which construction needs the midpoint of a side?	Median	Altitude	Angle bisector	Incircle	●○○○

Questions	A	B	C	D	A B C D
134. Which construction gives the centre of the inscribed circle?	Medians	Altitudes	Angle bisectors	Perpendicular bisectors	○○●○
135. Which construction gives the centre of the circle through 3 vertices?	Angle bisectors	Perpendicular bisectors	Medians	Altitudes	○●○○
136. How many circles pass through three non-collinear points?	Exactly one	Two	None	Infinitely many	●○○○
137. For an obtuse triangle, which pair of centres lies outside it?	Centroid and incentre	Incentre and circumcentre	Orthocentre and circumcentre	Centroid and orthocentre	○○●○
138. In isosceles △ABC (AB = AC), the four centres lie on	Side BC	Side AB	A single vertex only	The axis of symmetry from A	○○○●
139. Equilateral triangle of height 9 cm: centroid to base distance	6 cm	4.5 cm	2 cm	3 cm	○○○●
140. Equilateral triangle of side 6 cm: circumradius is	3 cm	3√3 cm	√3 cm	2√3 cm	○○○●
141. Equilateral triangle of side 6√3 cm: inradius is	6 cm	3 cm	9 cm	3√3 cm	○●○○
142. Right triangle 5, 12, 13 cm: inradius is	3 cm	2 cm	2.5 cm	4 cm	○●○○
143. Right triangle 5, 12, 13 cm: circumradius is	13 cm	6 cm	7.5 cm	6.5 cm	○○○●
144. Right triangle with legs 5 cm and 12 cm has area	60 cm²	17 cm²	65 cm²	30 cm²	○○○●
145. Triangle with sides 13, 14, 15 cm has area	168 cm²	84 cm²	42 cm²	90 cm²	○●○○
146. Altitudes, medians, bisectors and perpendicular bisectors give how many centres?	Four	Two	Three	One	●○○○
147. Midpoints of sides 6, 8, 10 cm are joined. New perimeter is	24 cm	6 cm	8 cm	12 cm	○○○●
148. A triangle with angles in the ratio 1 : 2 : 3 is	Equilateral	Obtuse	Right-angled	Acute equilateral	○○●○
149. Angles in ratio 2 : 3 : 4: the largest angle is	80°	90°	60°	40°	●○○○
150. Isosceles △ABC with AB = AC and ∠A = 40°: ∠B =	40°	80°	140°	70°	○○○●

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